

Introduction to Differential Equations

Notes taken by Runqiu Ye

Lectures by Gautum Iyer

Carnegie Mellon University

Spring 2026

Contents

1	Introduction	3
1.1	Introduction to ODEs	3
1.2	Autonomous linear systems	3
1.3	Time inhomogenous linear equations	5
2	Existence and Uniqueness	7

1 Introduction to ODEs

1.1 Example of ODEs

An example of ODE is as follows:

$$\dot{x} = -\alpha x,$$

where $\alpha > 0$. This represents exponential decay, whose solution is $x(t) = e^{-\alpha t}x_0$.

A more complicated example is the logistic equation where $P(t)$ represents the population density of some species, and we have

$$\dot{P} = rP \left(1 - \frac{P}{K}\right),$$

where $r, K > 0$. The solution is

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}}.$$

Another related example is Lotka-Volterra system. Let $x(t)$ be the population density of the rabbits and $y(t)$ is the population density of foxes. They can be described by the following model:

$$\begin{aligned}\dot{x} &= \alpha x - \beta xy, \\ \dot{y} &= -\gamma y + \delta xy,\end{aligned}$$

where the parameters $\alpha, \beta, \gamma, \delta > 0$.

1.2 Autonomous linear systems

Let $x : [0, \infty) \rightarrow \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$. Consider the equation $\dot{x} = Ax$. The solution to this equation is $x(t) = e^{tA}x_0$, where for any square matrix B we defined

$$e^B = \sum_{n=0}^{\infty} \frac{B^n}{n!}.$$

It is easy to check that this converges nicely, and thus we can take the derivative of e^{tA} against t term by term. We can then verify that $x(t) = e^{tA}x_0$ indeed solves the equation.

There are still something to say about this – how do we compute e^{tA} ? If $UAU^{-1} = D$ for some D diagonal, we then have $e^{At} = U^{-1}e^{Dt}U$. Now if the diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, we have $e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t})$.

However, if A is not diagonalizable, we have no option but to consider the Jordan normal form. Consider a basis change by setting $y = Ux$. We have

$$\dot{y} = U\dot{x} = UAx = UAU^{-1}y.$$

Therefore by choosing matrix U appropriately, we can guarantee UAU^{-1} is in Jordan normal form. Hence now we assume A is in Jordan normal form. Now decompose $\mathbb{R}^d$ into invariant subspaces $\{E_j\}$ such that $\mathbb{R}^d = \bigoplus_j E_j$ and each E_j has generalized eigenvalues $\lambda_j = a_j + ib_j$. There are two cases.

- If $b_j = 0$, there exists a basis B_j of E_j such that

$$(A)_{B_j} = \begin{bmatrix} a_j & 1 & & \\ & a_j & \ddots & \\ & & \ddots & 1 \\ & & & a_j \end{bmatrix}.$$

Then,

$$(e^{tA})_{B_j} = e^{ta_j} \sum_{k=0}^{m_j-1} \frac{t^k}{k!} ((A)_{B_j} - a_j I)^k,$$

where $m_j = \dim(E_j)$.

- If $b_j \neq 0$, there exists a basis B_j of E_j such that

$$(A)_{B_j} = \begin{bmatrix} C_j & I_2 & & \\ & C_j & \ddots & \\ & & \ddots & I_2 \\ & & & C_j \end{bmatrix},$$

where $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $C_j = \begin{bmatrix} a_j & -b_j \\ b_j & a_j \end{bmatrix}$. Then,

$$(e^{tA})_{B_j} = e^{a_j t} \begin{bmatrix} D_j(t) & & \\ & \ddots & \\ & & D_j(t) \end{bmatrix} \sum_{k=0}^{m_j-1} \frac{t^k}{k!} ((A)_{B_j} - \text{diag}(C_j, \dots, C_j))^k,$$

where $D_j(t) = \begin{bmatrix} \cos(b_j t) & -\sin(b_j t) \\ \sin(b_j t) & \cos(b_j t) \end{bmatrix}$.

From the case above, we can see that the long term behavior of the solution depends only on the real parts a_j . More concretely, let $E^\pm = \bigoplus_{j \in J^\pm} E_j$ where $J^\pm = \{j : \text{sgn}(a_j) = \pm 1\}$, and $E^0 \in \bigoplus_{j \in J^0} E_j$ where $J^0 = \{j : a_j = 0\}$. If $x_0 \in E^-$, then $x(t) \rightarrow 0$ as $t \rightarrow \infty$, and $|x(t)| \rightarrow \infty$ as $t \rightarrow -\infty$. This is the **stable subspace**. On the other hand, if $x_0 \in E^+$, then $|x(t)| \rightarrow \infty$ as $t \rightarrow \infty$, and $x(t) \rightarrow 0$ as $t \rightarrow -\infty$. This is the **unstable subspace**.

1.3 Time inhomogenous linear equations

Consider $\dot{x} = A(t)x$ where $A : \mathbb{R} \rightarrow \mathbb{R}^{d \times d}$. In the 1 dimensional case, we have $\dot{x} = a(t)x$. With $x(t_0) = x_0$, the solution to this equation is

$$x(t) = \exp \left(\int_{t_0}^t a(s) ds \right) x_0.$$

Theorem. Fix t_0, T . If for each $t_1, t_2 \in [t_0, T]$ we have $[A(t_1), A(t_2)] = 0$, then

$$x(t) = \exp \left(\int_{t_0}^t A(s) ds \right) x_0$$

solves $\dot{x} = A(t)x$ with $x(t_0) = x_0$.

Proof. Let $B(t) = \int_{t_0}^t A(s) ds$. Compute the derivative with the Wilcox Formula

$$\partial_t e^{B(t)} = \int_0^1 e^{\alpha B(t)} A(t) e^{(1-\alpha)B(t)} d\alpha. \quad (\text{W})$$

Note that (W) implies the theorem, because assumption implies that $[A(t), B(t)] = 0$ for all t . It then follows that $[A(t), e^{\alpha B(t)}] = 0$ and thus

$$\partial_t e^{B(t)} = \int_0^1 e^{B(t)} A(t) d\alpha = A(t) e^{B(t)}.$$

As for the proof for (W), the trick is to define $g(\alpha, t) = e^{-\alpha B(t)} \partial_t e^{\alpha B(t)}$. Then we compute

$$\begin{aligned} \partial_\alpha g &= -B e^{-\alpha B} \partial_t e^{\alpha B} + e^{-\alpha B} \partial_t (B e^{\alpha B}) \\ &= -B e^{-\alpha B} \partial_t e^{\alpha B} + e^{-\alpha B} A e^{\alpha B} + e^{-\alpha B} B \partial_t e^{\alpha B} \\ &= e^{-\alpha B} A e^{\alpha B}. \end{aligned}$$

Then, $g(1) - g(0) = \int_0^1 \partial_\alpha g d\alpha$. Therefore, $e^{-B} \partial_t e^B = \int_0^1 e^{-\alpha B} A e^{\alpha B} d\alpha$, and thus

$$\partial_t e^B = \int_0^1 e^{(1-\alpha)B} A e^{\alpha B} d\alpha,$$

as desired. □

Note that the equation $\dot{x} = A(t)x$ is linear, so any linear combination of solutions is again a solution. Say $\varphi_1, \dots, \varphi_d$ are d solutions, and $\varphi_i(t_0) = e_i$. Then, given $x \in \mathbb{R}^d$, $x(t) = \sum_{i=1}^d x_i \varphi_i(t)$, where $x_i = x_0 \cdot e_i$, solves the equation with initial condition x_0 . Now define

$$\Pi(t_0, t) = [\varphi_1(t) \quad \dots \quad \varphi_d(t)]$$

as the principle matrix. Note that $\partial_t \Pi(t_0, t) = A(t) \Pi(t_0, t)$, and $\Pi(t_0, t_0) = I$. Now the solution to $\dot{x} = A(t)x$ with $x(t_0) = x_0$ is given by $x(t) = \Pi(t, t_0)x_0$.

Proposition. $\Pi(s, t)\Pi(r, s) = \Pi(r, t)$.

Proof. Fix r, s and let $L(t) = \text{LHS}$ and $R(t) = \text{RHS}$. Note that $\partial_t L = A(t)\Pi(s, t)\Pi(r, s) = A(t)L(t)$. Similarly, $\partial_t R = A(t)R(t)$. For an initial data, note that $L(s) = R(s) = \Pi(r, s)$. $\square$

Corollary. $\Pi(s, t) = \Pi(t, s)^{-1}$.

Proposition (Liouville formula). $\partial_t(\det(\Pi(s, t))) = \text{tr}(A) \det(\Pi(s, t))$.

Proof. Exercise. $\square$

2 Existence and Uniqueness

Now we want to discuss existence and uniqueness of solution for $\dot{x} = f(t, x)$.

Theorem. Let $U \subset \mathbb{R}^{d+1}$ open, $f \in C(U, \mathbb{R}^d)$. Assume f locally Lipschitz. Then, given $(t_0, x_0) \in U$, there exists a locally unique solution to $\dot{x} = f(t, x)$ with $x(t_0) = x_0$.

By locally unique, we mean a solution in an interval around t_0 .